Sinais e Sistemas para Engenharia

MÓDULO 6 – TRANSFORMADA DE LAPLACE DIRETA E INVERSA

VERSÃO OCTAVE

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Laplace

A transformada de Laplace pode ser usada para resolver equações diferenciais. Além de ser uma alternativa muito eficiente quando comparada com a solução tradicional, o método de Laplace é particularmente vantajoso para sinais de entrada definidos por partes (em intervalos), para sinais periódicos, para sinais com descontinuidades ou para sinais definidos por impulsos.

A transformada de Laplace de um sinal $f(t)$ é definida por:

$$F(s) = \int_0^\infty f(t)e^{-st}dt$$

Notar que a integral acima é avaliada com respeito à variável t, no limites de integração de 0 a ∞ , de que modo que, ao final da integração, só restará a variável complexa "s" .

$f(t) - sinal\ original\ no\ tempo$
$F(s) - Transformada\ de\ Laplace\ do\ sinal\ f(t)$

Notar que "s" é uma variável complexa.

Passamos do domínio do tempo 't' (real) para o domínio de Laplace 's'(complexo).

Diz-se então que $F(s)$ é a transformada de Laplace de $f(t)$.

Transformada de Laplace e Transformada inversa de Laplace

A transformada de Laplace de um sinal $f(t)$ é definida por:

F(s) = $\int_0^\infty f(t)e^{-st}dt$

Que indicamos por:

$$f(t) \xRightarrow{\mathcal{L}} F(s)$$

$\mathcal{L}$ - *Transformada direta de Laplace*

$f(t) - sinal\ original\ no\ tempo$

$F(s) - Transformada\ de\ Laplace\ do\ sinal\ f(t)$

$$\boxed{F(s) = \int_0^\infty f(t)e^{-st}dt}$$

Esta transformação pode ser invertida de modo que:

$$F(s) \xRightarrow{\mathcal{L}^{-1}} f(t)$$

$\mathcal{L}^{-1}$ - *Transformada inversa de Laplace*

$F(s) - sinal\ no\ domínio\ da\ frequencia\ complexa.$

$f(t) - Transformada\ inversa\ de\ laplace\ do\ sinal\ F(s)$

$$\boxed{f(t) = \frac{1}{2\pi j}\oint_C F(s)e^{st}ds}$$

Exemplo 1 - $f(t) = u(t) = \begin{cases} 0; & t < 0 \\ 1; & t \geq 0 \end{cases}$

$$F(s) = \int_0^\infty f(t)e^{-st}dt = \int_0^\infty 1\, e^{-st}dt = -\frac{1}{s}e^{-st}\Big|_0^\infty = \frac{1}{s}$$

$$u(t) \xRightarrow{\mathcal{L}} \frac{1}{s}$$

Exemplo 2 - $f(t) = e^{-at}u(t)$

$$F(s) = \int_0^\infty f(t)e^{-st}dt = \int_0^\infty e^{-at}e^{-st}dt = \int_0^\infty e^{-(s+a)t}dt = -\frac{1}{(s+a)}e^{-(s+a)t}\Big|_0^\infty = \frac{1}{s+a}$$

$$e^{-at} \xRightarrow{\mathcal{L}} \frac{1}{s+a}$$

Exemplo 3 - $f(t) = \delta(t)$

$$F(s) = \int_0^{\infty} f(t)e^{-st}dt = \int_0^{\infty} \delta(t)e^{-st}dt = \int_{0^-}^{0^+} \delta(t)e^{-0}dt = 1$$

$$\delta(t) \xRightarrow{\mathcal{L}} 1$$

$$\cos(wt) = \frac{e^{jwt} + e^{-jwt}}{2}$$

Exemplo 4 - $f(t) = \cos(wt)\,u(t)$

$$F(s) = \int_0^{\infty} f(t)e^{-st}dt = \int_0^{\infty} \cos(wt)e^{-st}dt = \int_0^{\infty} \left(\frac{e^{jwt} + e^{-jwt}}{2}\right)e^{-st}dt = \frac{1}{2}\left(\frac{1}{s - jw} + \frac{1}{s + jw}\right) = \frac{s}{s^2 + w^2}$$

$$\cos(wt) \xRightarrow{\mathcal{L}} \frac{s}{s^2 + w^2}$$

Tabela Transformada de Laplace

$f(t);\ \ t \geq 0$	$F(s)$
Impulso Unitário $\delta(t)$	1
Degrau Unitário u(t)	$\frac{1}{s}$
t	$\frac{1}{s^2}$
$\frac{t^{n-1}}{(n-1)!}, (n = 1,2,3, \dots)$	$\frac{1}{s^n}$
$t^n, (n = 1,2,3, \dots)$	$\frac{n!}{s^{n+1}}$
e^{-at}	$\frac{1}{s+a}$
te^{-at}	$\frac{1}{(s+a)^2}$
$\frac{1}{(n-1)!} t^{n-1} e^{-at}, (n = 1,2,3, \dots)$	$\frac{1}{(s+a)^n}$
$t^n e^{-at}, (n = 1,2,3, \dots)$	$\frac{n!}{(s+a)^{n+1}}$
$sen\ \omega t$	$\frac{\omega}{s^2+\omega^2}$
$\cos \omega t$	$\frac{s}{s^2+\omega^2}$
$\operatorname{senh} \omega t$	$\frac{\omega}{s^2-\omega^2}$
$\cosh \omega t$	$\frac{s}{s^2-\omega^2}$
$\frac{1}{a}(1-e^{-at})$	$\frac{1}{s(s+a)}$
$\frac{1}{b-a}(e^{-at}-e^{-bt})$	$\frac{1}{(s+a)(s+b)}$
$\frac{1}{b-a}(be^{-bt}-ae^{-at})$	$\frac{s}{(s+a)(s+b)}$
$e^{-at} - ate^{-at}$	$\frac{s}{(s+a)^2}$
$\frac{Cs+D}{(s+\sigma)^2+w^2}$	$e^{-\sigma t}\left(C\cos(wt) + \frac{D-\sigma C}{w} sen(wt)\right)$
$\cos(at+b)$	$\frac{s\cos(b) - a\sin(b)}{s^2+a^2}$
$\sin(at+b)$	$\frac{s\sin(b) + a\cos(b)}{s^2+a^2}$
$\sqrt{t}$	$\frac{\sqrt{\pi}}{2s^{\frac{3}{2}}}$
$t^p, p > -1$	$\frac{\Gamma(p+1)}{s^{p+1}}$
$\frac{1}{ab}\left[1 + \frac{1}{a-b}(be^{-at} - ae^{-bt})\right]$	$\frac{1}{s(s+a)(s+b)}$
$\frac{1}{a^2}(1 - e^{-at} - ate^{-at})$	$\frac{1}{s(s+a)^2}$
$\frac{1}{a^2}(at - 1 + e^{-at})$	$\frac{1}{s^2(s+a)}$
$e^{-at} sen\ \omega t$	$\frac{\omega}{(s+a)^2+\omega^2}$
$e^{-at} \cos \omega t$	$\frac{s+a}{(s+a)^2+\omega^2}$
$\frac{\omega_n}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} sen\ \omega_n t\sqrt{1-\zeta^2}, (0 < \zeta < 1)$	$\frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$
$-\frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} sen\left(\omega_n t\sqrt{1-\zeta^2} - \phi\right)$ $\phi = tg^{-1}\frac{\sqrt{1-\zeta^2}}{\zeta}, \left(0 < \zeta < 1, 0 < \phi < \frac{\pi}{2}\right)$	$\frac{s}{s^2+2\zeta\omega_n s+\omega_n^2}$
$1 - \frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} sen\left(\omega_n t\sqrt{1-\zeta^2} + \phi\right)$ $\phi = \frac{tg^{-1}\sqrt{1-\zeta^2}}{\zeta}, \left(0 < \zeta < 1, 0 < \phi < \frac{\pi}{2}\right)$	$\frac{\omega_n^2}{s(s^2+s\zeta\omega_n s+\omega_n^2)}$
$1 - \cos \omega t$	$\frac{\omega^2}{s(s^2+\omega^2)}$
$\omega t - sen\ \omega t$	$\frac{\omega^3}{s^2(s^2+\omega^2)}$
$sen\ \omega t - \omega t \cos \omega t$	$\frac{2\omega^3}{(s^2+\omega^2)^2}$
$\frac{1}{2\omega} t\ sen\ \omega t$	$\frac{s}{(s^2+\omega^2)^2}$
$t \cos \omega t$	$\frac{s^2-\omega^2}{(s^2+\omega^2)^2}$
$\frac{1}{\omega_2^2-\omega_1^2}(\cos \omega_1 t - \cos \omega_2 t), (\omega_1^2 \neq \omega_2^2)$	$\frac{s}{(s^2+\omega_1^2)(s^2+\omega_2^2)}$
$\frac{1}{2\omega}(sen\ \omega t + \omega t \cos \omega t)$	$\frac{s^2}{(s^2+\omega^2)^2}$

Propriedades da Transformada de Laplace

1) Linearidade

$$f(t) \xRightarrow{\mathcal{L}} F(s)$$

$$g(t) \xRightarrow{\mathcal{L}} G(s)$$

$$af(t) + bg(t) \xRightarrow{\mathcal{L}} aF(s) + bG(s)$$

2) Derivada no Tempo

$$f(t) \xRightarrow{\mathcal{L}} F(s)$$

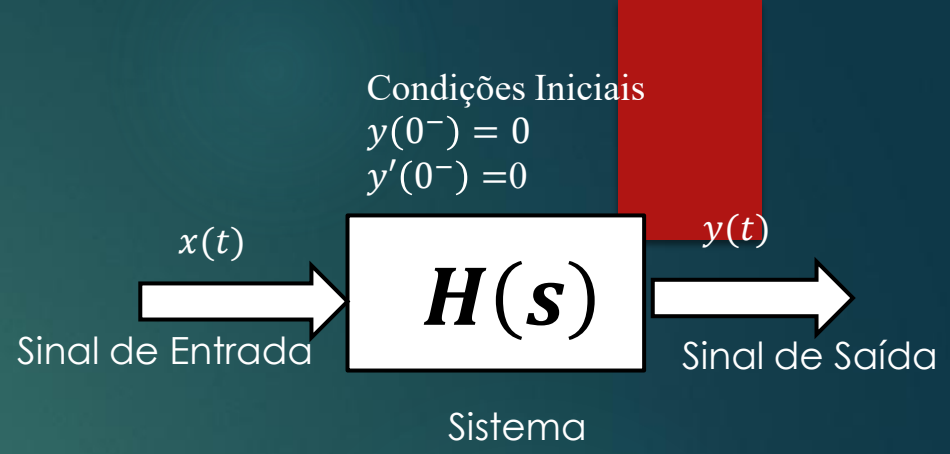
$$f'(t) \xRightarrow{\mathcal{L}} sF(s) - f(0)$$

$$f''(t) \xRightarrow{\mathcal{L}} s^2F(s) - sf(0) - f'(0)$$

$$f'''(t) \xRightarrow{\mathcal{L}} s^3F(s) - s^2f(0) - sf'(0) - f''(0)$$

Aplicação das Propriedades 1 e 2:

Transformada de Laplace de Equações Diferenciais



$$se \quad x(t) \xRightarrow{\mathcal{L}} X(s)$$

$$e \quad y(t) \xRightarrow{\mathcal{L}} Y(s)$$

$$y''(t) + a_1\, y'(t) + a_2\, y(t) = b_1\, x(t) \quad \xRightarrow{\mathcal{L}} \quad s^2 Y(s) + a_1\, sY(s) + a_2\, Y(s) = b_1\, X(s)$$

Função de Transferência

$$s^2 Y(s) + a_1\, sY(s) + a_2\, Y(s) = b_1\, X(s)$$

$$Y(s)(s^2 + a_1\, s + a_2) = b_1\, X(s)$$

$$\frac{Y(s)}{X(s)} = \frac{b_1}{s^2 + a_1\, s + a_2}$$

$$H(s) = \frac{Y(s)}{X(s)} = \frac{b_1}{s^2 + a_1\, s + a_2}$$

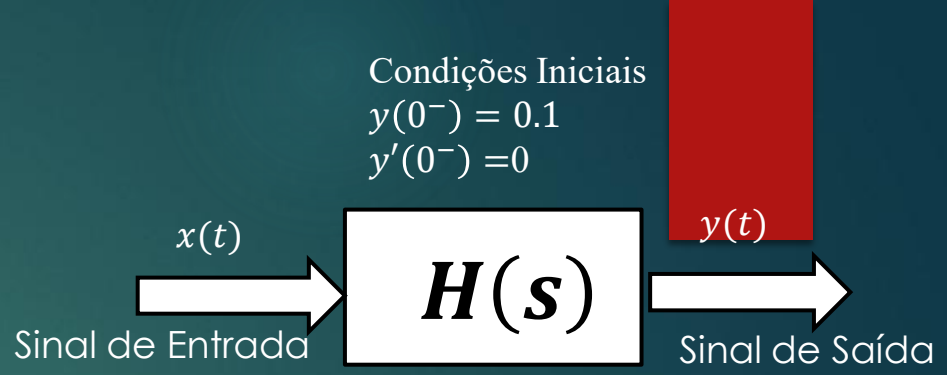
Aplicação das Propriedades 1 e 2:

Transformada de Laplace de Equações Diferenciais

$$se \quad x(t) \xRightarrow{\mathcal{L}} X(s)$$

$$e \quad y(t) \xRightarrow{\mathcal{L}} Y(s)$$

A entrada $x(t)$ está implicitamente multiplicada por $u(t)$, isto é, a entrada começa a ser aplicada no tempo $t = 0$, o que justifica assumirmos condições iniciais nulas na entrada: $x(0) = 0 \;\; e \;\; x'(0) = 0$.



$$H(s) = \frac{6s + 45}{s^2 + 6s + 45}$$

$$\boldsymbol{y''(t) + 6y'(t) + 45y(t) = 6x'(t)} + 45x(t) \xRightarrow{\mathcal{L}} (s^2Y(s) - 0.1s - 0) + 6(sY(s) - 0.1) + 45Y(s) = 6sX(s) + 45X(s)$$

Função de Transferência - Condições iniciais nulas

$$H(s) = \frac{Y(s)}{X(s)}$$

$$s^2Y(s) + 6sY(s) + 45Y(s) = 6sX(s) + 45X(s)$$

$$Y(s)(s^2 + 6s + 45) = (6s + 45)X(s)$$

$$H(s) = \frac{6s + 45}{s^2 + 6s + 45}$$

Propriedades da Transformada de Laplace

3) Integração no Tempo

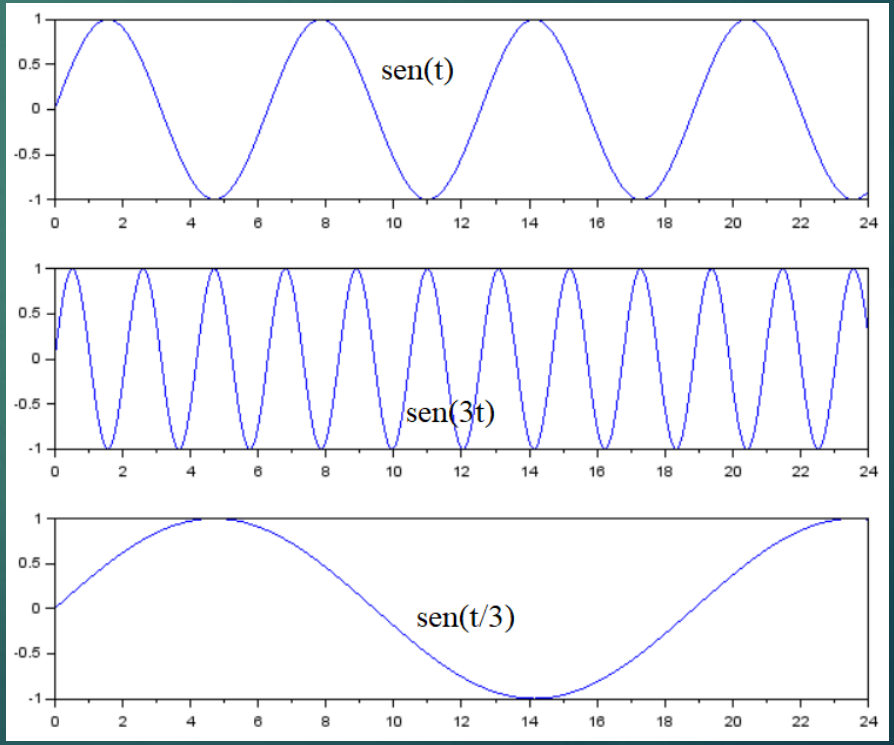
$$f(t) \xRightarrow{\mathcal{L}} F(s)$$

$$\int_0^t f(\tau)d\tau \xRightarrow{\mathcal{L}} \frac{1}{s}F(s)$$

4) Escala no tempo

$$f(t) \xRightarrow{\mathcal{L}} F(s)$$

$$f(at) \xRightarrow{\mathcal{L}} \frac{1}{a}F\left(\frac{s}{a}\right)$$

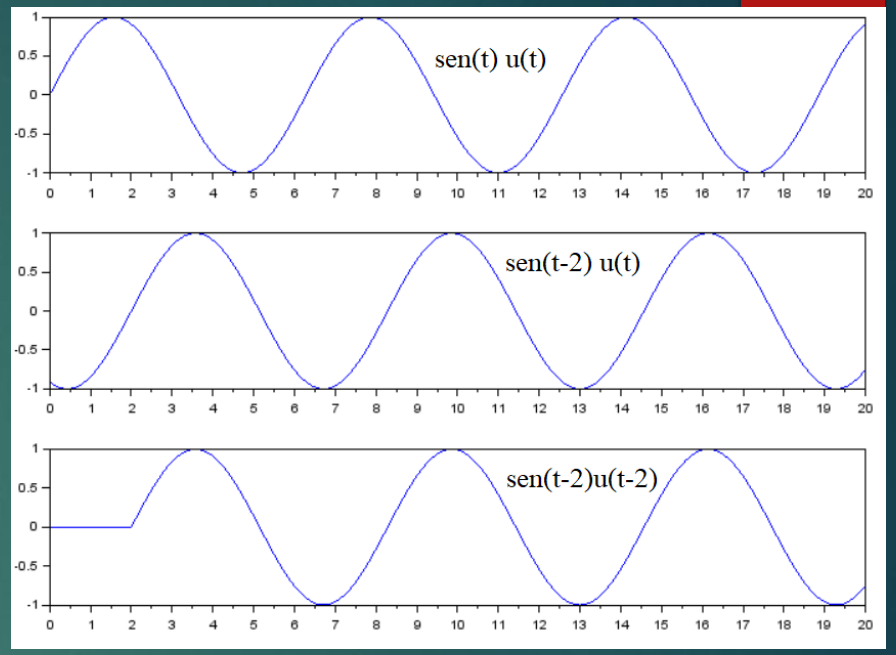


Propriedades da Transformada de Laplace

5) Deslocamento no tempo

$$f(t) \xRightarrow{\mathcal{L}} F(s)$$

$$f(t-a)u(t-a) \xRightarrow{\mathcal{L}} e^{-as}F(s)$$



6) Deslocamento em frequência

$$f(t) \xRightarrow{\mathcal{L}} F(s)$$

$$e^{-at}f(t) \xRightarrow{\mathcal{L}} F(s+a)$$

Propriedades da Transformada de Laplace

7) Modulação por cosseno

$$f(t) \xRightarrow{\mathcal{L}} F(s)$$

$$f(t)\cos(w_o t) \xRightarrow{\mathcal{L}} \frac{F(s+jw_o)+F(s-jw_o)}{2}$$

8) Modulação por seno

$$f(t) \xRightarrow{\mathcal{L}} F(s)$$

$$f(t)\mathrm{sen}(w_o t) \xRightarrow{\mathcal{L}} \frac{F(s+jw_o)-\ F(s-jw_o)}{2j}$$

Propriedades da Transformada de Laplace

9) Valor Inicial

$$f(t) \xRightarrow{\mathcal{L}} F(s)$$

$$f(0^+) = \lim_{s \to \infty} sF(s)$$

$$f(t) \xRightarrow{\mathcal{L}} F(s) = \frac{3s+4}{5s^2+8s+22}$$

$f(0^+)$=?

$$f(0^+) = \lim_{s \to \infty} sF(s) = \lim_{s \to \infty}\left(s\frac{3s+4}{5s^2+8s+22}\right) = \lim_{s \to \infty}\left(\frac{3s^2+4s}{5s^2+8s+22}\right) = \frac{3}{5}$$

10) Valor Final

$$f(t) \xRightarrow{\mathcal{L}} F(s)$$

$$\lim_{t \to \infty} f(t) = \lim_{s \to 0} F(s)$$

$$f(t) \xRightarrow{\mathcal{L}} F(s) = \frac{3s+4}{5s^2+8s+22}$$

$\lim_{t \to \infty} f(t)$ =?

$$\lim_{t \to \infty} f(t) = \lim_{s \to 0} F(s) = \lim_{s \to 0}\left(\frac{3s+4}{5s^2+8s+22}\right) = \frac{4}{22} = \frac{2}{11}$$

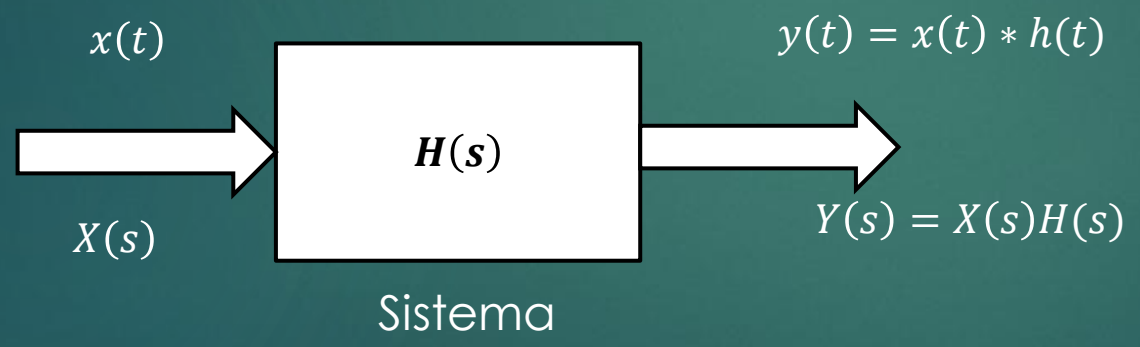
Propriedades da Transformada de Laplace

11) Convolução no Tempo

$$x(t) \xRightarrow{\mathcal{L}} X(s)$$

$$h(t) \xRightarrow{\mathcal{L}} H(s)$$

$$x(t) * h(t) \xRightarrow{\mathcal{L}} X(s)H(s)$$



Transformada Inversa de Laplace

$$f(t) \xrightarrow{Transformada\ Direta} F(s)$$

$$F(s) = \int_0^\infty f(t)e^{-st}dt$$

$$F(s) \xrightarrow{Transformda\ Inversa} f(t)$$

$$f(t) = \frac{1}{2\pi j}\oint_C F(s)e^{st}ds$$

Para cálculo da transformada inversa de Laplace, devemos avaliar uma integral de contorno da função complexa ($F(s)e^{st}$), em um contorno que contenha os polos da função. Esta integral pode ser avaliada então pelo Teorema de Cauchy:

$$f(t) = \frac{1}{2\pi j}\oint_C (F(s)e^{st})ds = \sum_{k=1}^{n} R_k$$

Onde:
C é um contorno fechado no plano s, que circunscreva todos os n polos p_k de $F(s)e^{st}$
R_k - é o resíduo do polo p_k de $F(s)e^{st}$)
n = número de polos de $F(s)e^{st}$

Cálculo da transformada inversa de Laplace

- Serão analisados 3 métodos

Método 1 – Expansão em Frações Parciais de $H(s)$ e Inspeção na Tabela.

Método 2 – Soma dos Resíduos dos polos de $H(s)e^{st}$

Método 3 – Sistema de Equações e Inspeção na Tabela

Frações Parciais e Resíduos de polos

Sistemas lineares e invariantes no tempo são normalmente representados por uma função racional complexa, isto é, pelo quociente de dois polinômios de uma variável complexa "s". Assim:

$$H(s) = \frac{P(s)}{Q(s)}$$

$$\boxed{\mathrm{H}(s) \xrightarrow{Transformda\ Inversa} h(t)}$$

Onde
H(s) é uma função racional complexa
s é uma variável complexa
P(s) e Q(s) são funções polinomiais complexas

A função h(t) no tempo, que é a transformada inversa de Laplace de H(s), pode ser avaliada pela soma dos resíduos dos polos de $(H(s)e^{st})$. A função h(t) também pode ser encontrada a partir da expansão em frações parciais de H(s) e consulta a tabelas.

Normalmente a ordem de Q(s) é maior que a ordem de P(s). Caso a ordem de P(s) seja maior ou igual à ordem de Q(s) dizemos que a função H(s) é imprópria, que pode ser separada em uma função polinomial somada a uma função racional própria.

Caso 1)
Denominador com polos simples

$$H(s) = \frac{3}{s^2+5s+6} = \frac{3}{(s+2)(s+3)}$$

(polos -2 e -3)

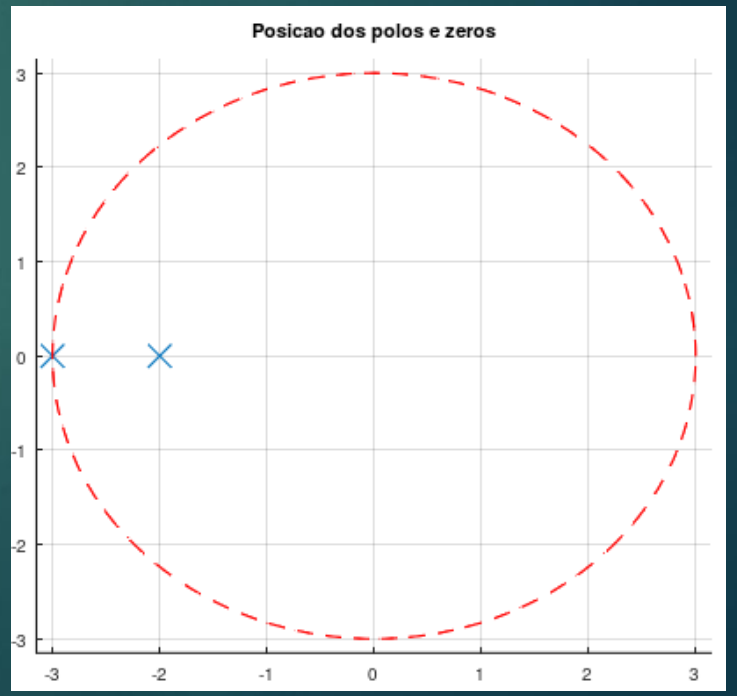
```
>> pkg load control
>> s=tf('s');
```

```
>> Hs=3/(s^2+5*s+6)

Transfer function 'Hs' from input 'u1' to output
..

             3
 y1:  -------------
      s^2 + 5 s + 6

Continuous-time model.
>> plot_polos_zeros(Hs)
```



Método 1 – Expansão em Frações Parciais de H(s) e Inspeção na Tabela.

$$H(s) = \frac{3}{(s+2)(s+3)} = \frac{A}{(s+2)} + \frac{B}{(s+3)}$$

$$A = H(s)\,(s+2)\Big|_{s=-2} = \frac{3}{\cancel{(s+2)}(s+3)}\cancel{(s+2)}\Big|_{s=-2} = \frac{3}{(s+3)}\Big|_{s=-2} = \frac{3}{(-2+3)} = 3$$

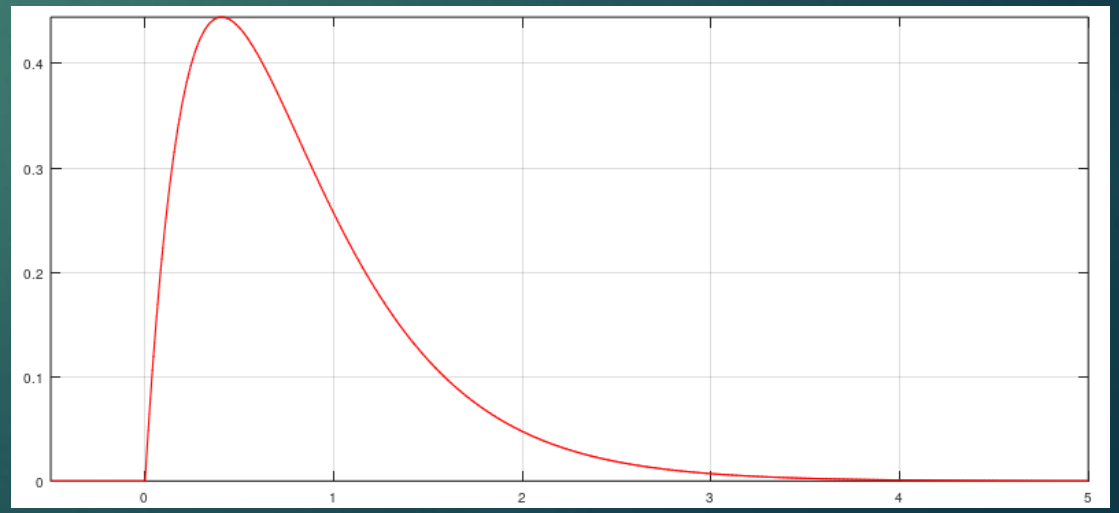
$$B = H(s)\,(s+3)\Big|_{s=-3} = \frac{3}{(s+2)\cancel{(s+3)}}\cancel{(s+3)}\Big|_{s=-3} = \frac{3}{(s+2)}\Big|_{s=-3} = \frac{3}{(-3+2)} = -3$$

$$H(s) = \frac{A}{(s+2)} + \frac{B}{(s+3)} = \frac{3}{(s+2)} + \frac{-3}{(s+3)}$$

e^{-at}	$\frac{1}{s+a}$

$$H(s) \xRightarrow{\mathcal{L}^{-1}} h(t) = [3e^{-2t} - 3e^{-3t}]\mathrm{u(t)}$$

```
>> h = @(t) (3*exp(-2*t)-3*exp(-3*t)).*f_u(t);
>> ct=1/2;
>> t=linspace(-ct,10*ct,1001);
>> plot(t,h(t),'r');
```



```
function ft=ilaplace_fga(H,compact)
    [num_c,den_c]=tfdata(H,'v');
    [r, p, k, e] = residue (num_c, den_c);
    s=tf('s');
    cmpx=true;
    f_deff="ft=@(t)(";

    if (!isempty(k)) % fracao impropia
        printf("f_total(t)=")
        k=flip(k);
        for (i=1:length(k))
            if k(i)!=0
                strd=sprintf("(%.2f)*f_d",k(i));
                if i>1
                    for j=1:i-1
                        strd=strcat(strd ,"'");
                    end
                end
                strd=strcat(strd , "(t)");
                printf("%s +",strd)
            end
        end
        printf(" ft(t)\n")
    end

    for n=1:length(r)
      if (abs(imag(p(n)))<1e-8)
        if (e(n)==1)
            if(abs(r(n))>1e-8)
                if(abs(p(n))<1e-5)
                    str=sprintf("%.5f",r(n));
                else
                    str=sprintf("%.5f*exp(%.5f*t)",r(n),p(n));
                end
                f_deff=strcat(f_deff,"+(",str,")");
            end
        elseif (e(n)==2)
            if(abs(r(n))>1e-8)
                if(abs(p(n))<1e-5)
```

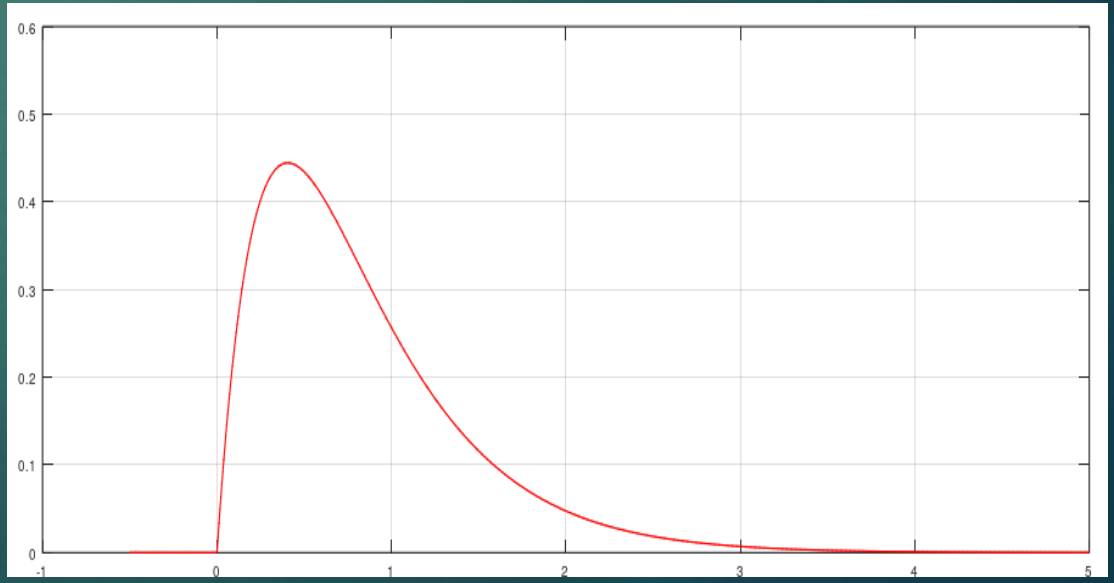
$$H(s) = \frac{3}{(s+2)(s+3)} = \frac{A}{(s+2)} + \frac{B}{(s+3)}$$

$$H(s) \xRightarrow{\mathcal{L}^{-1}} h(t) = [3e^{-2t} - 3e^{-3t}]u(t)$$

```
>> Hs=3/(s^2+5*s+6);
>> ft = ilaplace_fga(Hs,true);
ft=@(t)(+(-3.00000*exp(-3.00000*t))+(3.00000*exp(-2.00000*t)) ).*f_u(t);
>>
>> ct=1/2;
>> t=linspace(-ct,10*ct,1001);
>> plot(t,ft(t),'r');
>> grid on;
```

• • •

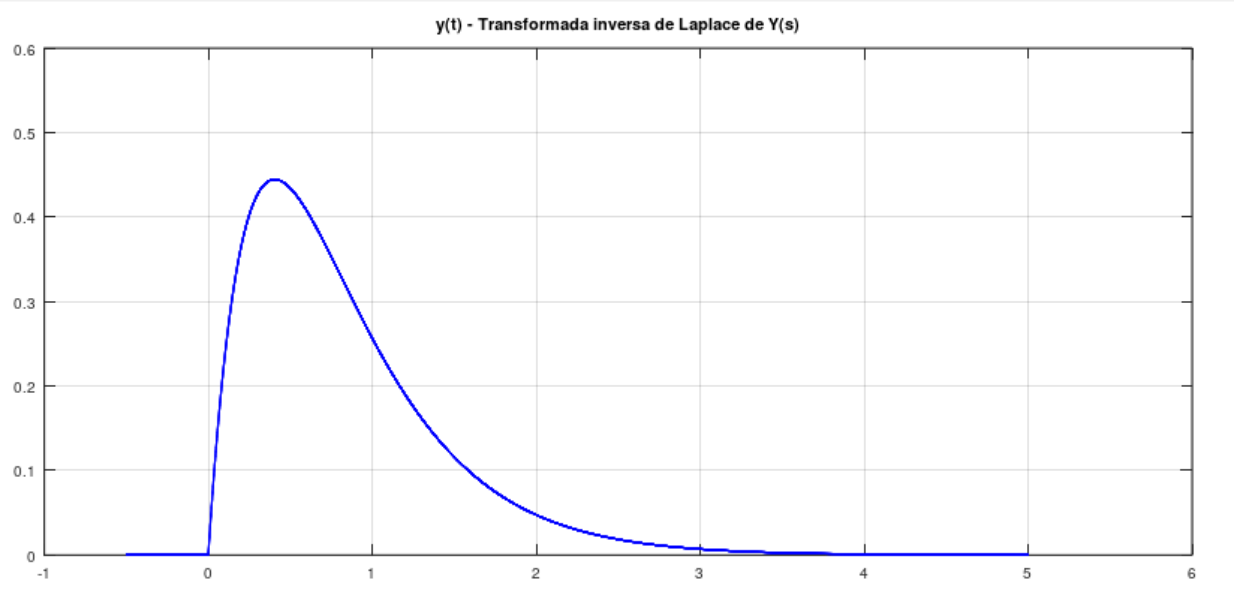
```
                end;
            endif
        end
    end
    f_deff=strcat(f_deff," ).*f_u(t);");
    disp(f_deff)
    eval(f_deff)
endfunction
```



```
function [yo,t]=Plot_Transformada_Inversa(Hs,L,cor)
    Hs=minreal(Hs);
    [num_c,den_c]=tfdata(Hs,'v');
    [r, p, k, e] = residue (num_c, den_c);
    ft=ilaplace_fga(Hs,true);
    index=find(abs(p)>0);
    min_p=0;
    if( !isempty(index) )
        min_p = min(abs(p(index)));
    end
    if(min_p>0)
       ct = 1/min_p; %constante de tempo
       max_t = 10 * L * ct;
    else
       ct=0.1;
       max_t = 10*ct;
    end
    plot_impulse=false;
    t=linspace(-ct,max_t,1001);;
    hold on; grid on;
    if (num_c==[1] && den_c==[1])
      clf;
       yo=f_d(t);
       plot(t,yo,cor);
       plot_arrow_fga([0,0],[0,1],cor)
    else
      if isequal(axis, [-0.2000   1.0000  -1.0000   1.0000] )
         clf;
         plot_impulse=true;
      end;
      p=abs(p);
      ws = 50*max(abs(p));
      T=2* pi /ws;
      min_p=0;
      yo=ft(t);
      plot(t,yo,cor);
      title("y(t) - Transformada inversa de Laplace de Y(s)");
      hold on; grid on;
      if (!isempty(k)) % fracao impropia
          k=flip(k);
          I=max(abs(k));
          if length(k)>=1
              plot([0,0],[0,max(yo)*k(1)/I]);
              plot_arrow_fga([0,0],[0,max(yo)*k(1)/I],cor);
          end
          if length(k)>=2
              plot([-dt/100,-dt/100],
                   [-max(yo)/2*k(2)/I,max(yo)/2*k(2)/I]);
              plot_darrow_fga([-dt/100,-dt/100],
                   [-max(yo)/2*k(2)/I,max(yo)/2*k(2)/I],cor);
          end
          if (plot_impulse)
              plot_arrow_fga([0,0],[0,max(yo)/I],'r')
          end
      else
          if (plot_impulse)
              plot_arrow_fga([0,0],[0,3/4*max(yo)],'r')
          end
      end
    end
endfunction
```

$$H(s) = \frac{3}{(s+2)(s+3)} \xRightarrow{\mathcal{L}^{-1}} h(t) = [3e^{-2t} - 3e^{-3t}]u(t)$$

```
>> Hs=3/(s^2+5*s+6);
>> [yo,t]=Plot_Transformada_Inversa(Hs,1,'b');
ft=@(t)(+(-3.00000*exp(-3.00000*t))+(3.00000*exp(-2.00000*t)) ).*f_u(t);
```



Método 2 – Resíduos dos Polos de $H(s)e^{st}$

$$H(s) = \frac{3}{(s+2)(s+3)}$$

$$h(t) = \frac{1}{2\pi j}\oint_C (H(s)e^{st})ds = \sum_{k=1}^{n} R_k$$

$$R_1 = (H(s)e^{st})(s+2)\Big|_{s=-2} = \frac{3e^{st}}{\cancel{(s+2)}(s+3)}\cancel{(s+2)}\Big|_{s=-2} = \frac{3e^{st}}{(s+3)}\Big|_{s=-2} = \frac{3e^{-2t}}{(-2+3)} = 3e^{-2t}$$

$$R_2 = (H(s)e^{st})(s+3)\Big|_{s=-3} = \frac{3e^{st}}{(s+2)\cancel{(s+3)}}\cancel{(s+3)}\Big|_{s=-3} = \frac{3e^{st}}{(s+2)}\Big|_{s=-3} = \frac{3e^{-3t}}{(-3+2)} = -3e^{-3t}$$

$$H(s) \xRightarrow{\mathcal{L}^{-1}} h(t) = \sum_{k=1}^{n} R_k = [3e^{-2t} - 3e^{-3t}]\mathrm{u(t)}$$

Método 3 – Sistema de equações e inspeção na tabela

$$H(s) = \frac{3}{(s+2)(s+3)} = \frac{A}{(s+2)} + \frac{B}{(s+3)}$$

Multiplicar ambos os lados por $[(s+2)(s+3)]$

$$[(s+2)(s+3)]\frac{3}{[(s+2)(s+3)]} = [(s+2)(s+3)]\left[\frac{A}{(s+2)} + \frac{B}{(s+3)}\right]$$

$$3 = A(s+3) + B(s+2)$$

$$3 = (A+B)s + (3A+2B)$$

$$\begin{cases} A + B = 0 \\ 3A + 2B = 3 \end{cases}$$

Resolvendo simultaneamente temos que $A = 3\ e\ B = -3$

$$H(s) = \frac{A}{(s+2)} + \frac{B}{(s+3)} = \frac{3}{(s+2)} + \frac{-3}{(s+3)}$$

e^{-at}	$\frac{1}{s+a}$

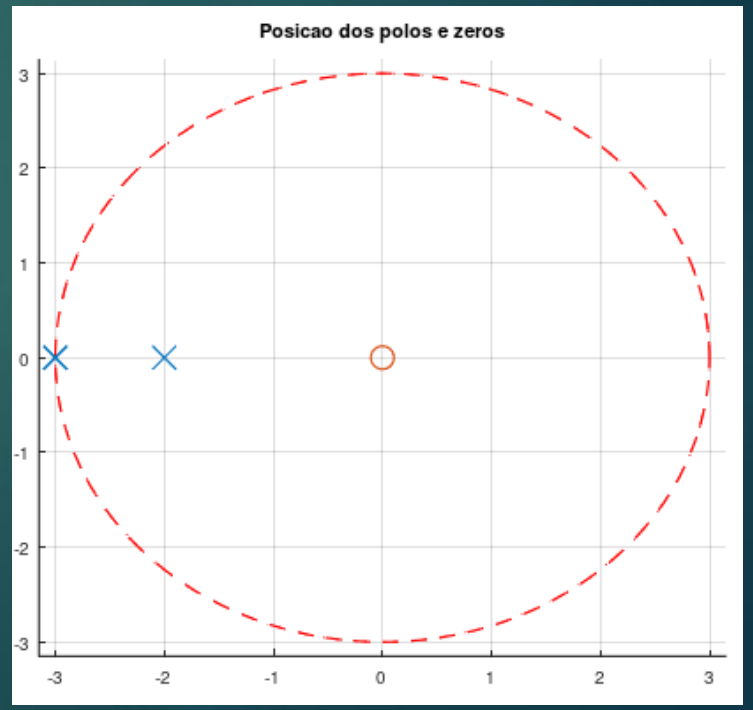
$$H(s) \xRightarrow{\mathcal{L}^{-1}} h(t) = [3e^{-2t} - 3e^{-3t}]u(t)$$

Caso 2) Denominador com polos duplos

$$H(s) = \frac{5s}{s^3+8s^2+21s+18} = \frac{5s}{(s+2)(s+3)^2}$$

(polos -2, -3 e -3)
(zero em 0)

```
>> Hs=5*s/((s+2)*(s+3)^2);
>> plot_polos_zeros(Hs)
```



Método 1 – Expansão em Frações Parciais de H(s) e Inspeção na Tabela.

$$H(s) = \frac{5s}{(s+2)(s+3)^2} = \frac{A}{(s+2)} + \frac{B}{(s+3)^2} + \frac{C}{(s+3)}$$

Notar que, na Expansão em Fações Parciais de um polo duplo e um simples há 3 termos

$$A = H(s)\,(s+2)\Big|_{s=-2} = \frac{5s}{\cancel{(s+2)}(s+3)^2}\cancel{(s+2)}\Big|_{s=-2} = \frac{5s}{(s+3)^2}\Big|_{s=-2} = \frac{5(-2)}{(-2+3)^2} = -10$$

$$B = H(s)(s+3)^2\Big|_{s=-3} = \frac{5s}{(s+2)\cancel{(s+3)^2}}\cancel{(s+3)^2}\Big|_{s=-3} = \frac{5s}{(s+2)}\Big|_{s=-3} = \frac{5(-3)}{(-3+2)} = 15$$

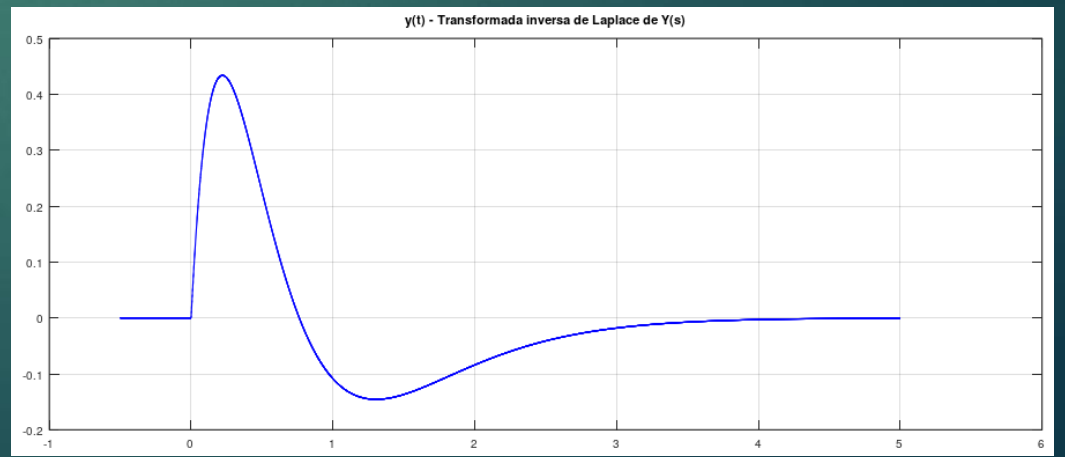
$$C = \frac{\partial}{\partial s}[H(s)(s+3)^2]\Big|_{s=-3} = \frac{\partial}{\partial s}\left[\frac{5s}{(s+2)}\right]\Big|_{s=-3} = \frac{10}{(s+2)^2}\Big|_{s=-3} = 10$$

```
>> Hs=5*s/((s+2)*(s+3)^2);
>> [yo,t]=Plot_Transformada_Inversa(Hs,1,'b');
ft=@(t)(+(10.00000*exp(-3.00000*t))+(15.00000*t.*exp(-3.00000*t)
)+(-10.00000*exp(-2.00000*t)) ).*f_u(t);
```

$$H(s) = \frac{A}{(s+2)} + \frac{B}{(s+3)^2} + \frac{C}{(s+3)} = \frac{-10}{(s+2)} + \frac{15}{(s+3)^2} + \frac{10}{(s+3)}$$

e^{-at}	$\frac{1}{s+a}$
te^{-at}	$\frac{1}{(s+a)^2}$

$$H(s) \xRightarrow{\mathcal{L}^{-1}} h(t) = [-10e^{-2t} + 15te^{-3t} + 10e^{-3t}]\mathrm{u(t)}$$



Método 2 – Resíduos dos Polos de $H(s)e^{st}$

$$H(s) = \frac{5s}{(s+2)(s+3)^2}$$

Notar que, na Expansão em Fações Parciais de um polo duplo e um simples há 3 termos, mas há somente 2 resíduos!!

$$h(t) = \frac{1}{2\pi j}\oint_C (H(s)e^{st})ds = \sum_{k=1}^{n} R_k$$

$$R_1 = H(s)e^{st}(s+2)\dashv|_{s=-2} = \frac{5se^{st}}{\cancel{(s+2)}(s+3)^2}\cancel{(s+2)}\Big|_{s=-2} = \frac{5se^{st}}{(s+3)^2}\Big|_{s=-2} = \frac{5(-2)e^{-2t}}{(-2+3)^2} = -10e^{-2t}$$

$$R_2 = \frac{\partial}{\partial s}[H(s)e^{st}(s+3)^2]\Big|_{s=-3} = \frac{\partial}{\partial s}\left[\frac{5se^{st}}{\cancel{(s+2)}(s+3)^2}\cancel{(s+3)^2}\right]\Big|_{s=-3} = \frac{\partial}{\partial s}\left[\frac{5se^{st}}{(s+2)}\right]\Big|_{s=-3} = 15te^{-3t} + 10e^{-3t}$$

$$H(s) \xRightarrow{\mathcal{L}^{-1}} h(t) = \sum_{k=1}^{n} R_k = [-10e^{-2t} + 15te^{-3t} + 10e^{-3t}]\mathrm{u(t)}$$

Método 3 – Sistema de equações e inspeção na tabela

$$H(s) = \frac{5s}{(s+2)(s+3)^2} = \frac{A}{(s+2)} + \frac{B}{(s+3)^2} + \frac{C}{(s+3)}$$

$$[(s+2)(s+3)^2]\frac{5s}{(s+2)(s+3)^2} = [(s+2)(s+3)^2]\left[\frac{A}{(s+2)} + \frac{B}{(s+3)^2} + \frac{C}{(s+3)}\right]$$

$$5s = A(s+3)^2 + B(s+2) + C(s+3)(s+2) = A(s^2+6s+9) + B(s+2) + C(s^2+5s+6)$$

$$5s = (A+C)s^2 + (6A+B+5C])s + (9A+2B+6C)$$

$$\begin{cases} A + \quad C = 0 \\ 6A + B + 5C = 5 \\ 9A + 2B + 6C = 0 \end{cases}$$

Resolvendo simultaneamente temos que $A = -10$, $B = 15$ e $C = 10$

$$H(s) = \frac{A}{(s+2)} + \frac{B}{(s+3)^2} + \frac{C}{(s+3)} = \frac{-10}{(s+2)} + \frac{15}{(s+3)^2} + \frac{10}{(s+3)}$$

$$H(s) \xRightarrow{\mathcal{L}^{-1}} h(t) = [-10e^{-2t} + 15te^{-3t} + 10e^{-3t}]u(t)$$

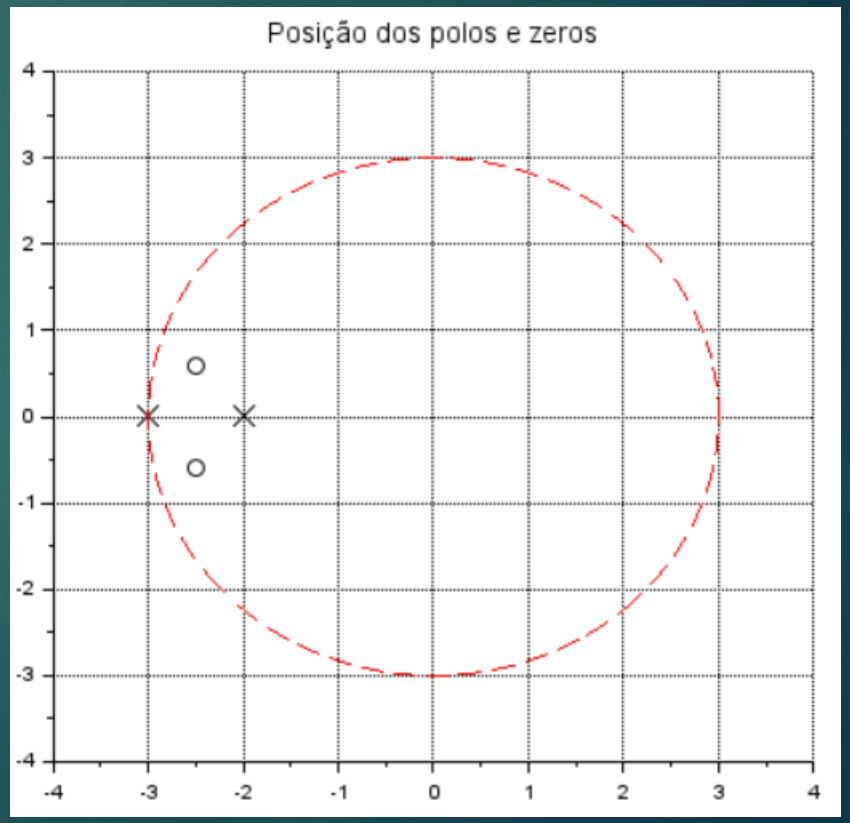
Caso 3) Função racional imprópria

$$H(s) = \frac{5s^2 + 25s + 33}{s^2 + 5s + 6}$$

(polos -2, -3)

```
--> Hs=(5*s^2+25*s+33)/(s^2+5*s+6);

--> plot_polos_zeros(Hs)
```



Método 1 – Expansão em Frações Parciais de H(s) e Inspeção na Tabela.

$$H(s) = \frac{5s^2 + 25s + 33}{s^2 + 5s + 6} = 5 + \frac{3}{s^2 + 5s + 6} = 5 + \frac{3}{(s+3)(s+2)} = 5 + \frac{A}{(s+2)} + \frac{B}{(s+3)}$$

$$\begin{array}{r|l} 5s^2 + 25s + 33 & s^2 + 5s + 6 \\ \cline{2-2} 5s^2 + 25s + 30 & 5 \\ \hline 3 & \end{array}$$

```
>> Hs=(5*s^2+25*s+33)/(s^2+5*s+6);
>> [yo,t]=Plot_Transformada_Inversa(Hs,1,'b');
f_total(t)=(5.00)*f_d(t) + ft(t)
ft=@(t)(+(-3.00000*exp(-3.00000*t))+(3.00000*exp(-2.00000*t)) ).*f_u(t);
```

$$A = \left[\frac{3}{(s+3)(s+2)}\right](s+2)\Big|_{s=-2} = 3$$

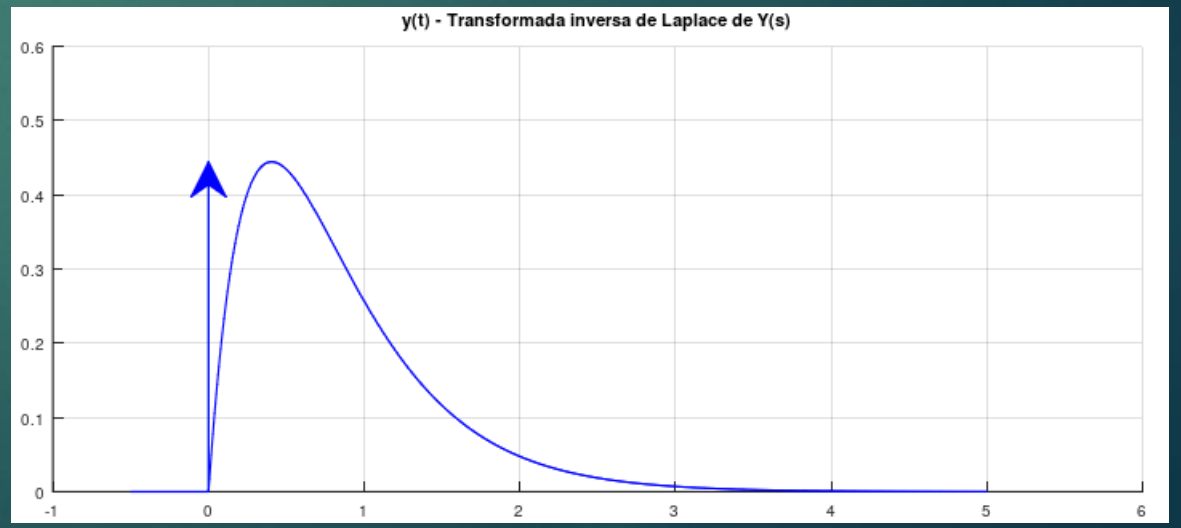
$$B = \left[\frac{3}{(s+3)(s+2)}\right](s+3)\Big|_{s=-3} = -3$$

$$H(s) = 5 + \frac{3}{(s+2)} + \frac{-3}{(s+3)}$$

Impulso Unitário $\delta(t)$	1

e^{-at}	$\frac{1}{s+a}$

$$H(s) \xrightarrow{\mathcal{L}^{-1}} h(t) = 5\delta(t) + [3e^{-2t} - 3e^{-3t}]u(t)$$

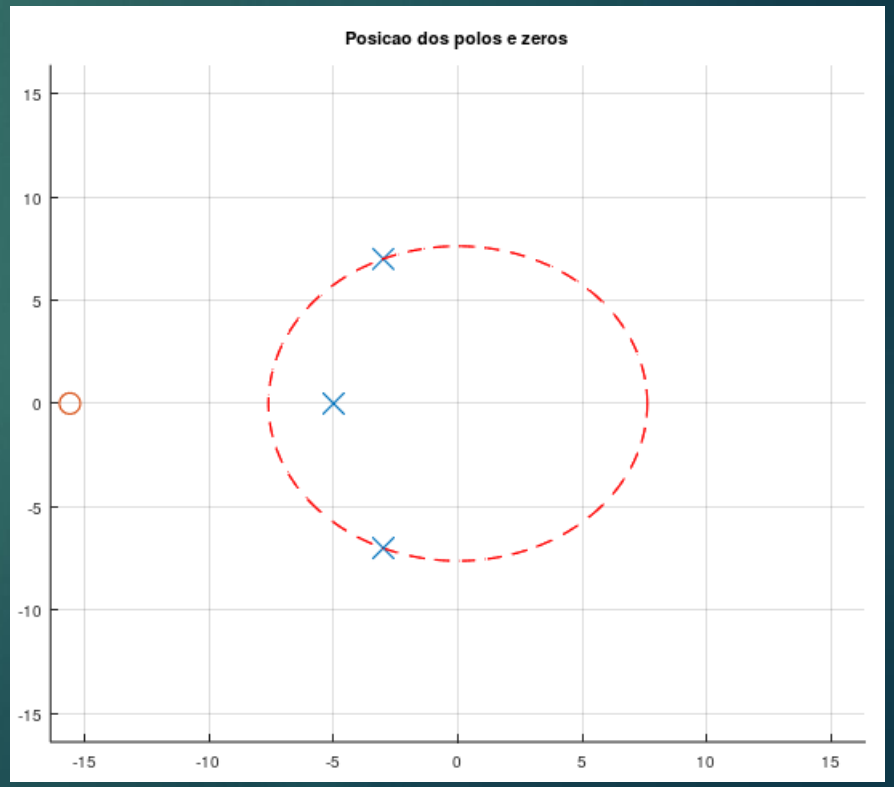


Caso 4A) Polos complexos conjugados

$$H(s) = \frac{(10s + 156)}{(s+5)(s^2 + 6s + 58)} = \frac{(10s + 156)}{(s+5)\left((s+3)^2 + 7^2\right)}$$

(polos -5, -3+j7, -3-j7)
(zero -15.6)

```
>> Hs=(10*s+156)/((s+5)*(s^2+6*s+58));
>> plot_polos_zeros(Hs)
```



Método 1 – Expansão em Frações Parciais de H(s) e Inspeção na Tabela.

$$H(s) = \frac{(10s+156)}{(s+5)\left((s+3)^2+7^2\right)} = \frac{(10s+156)}{(s+5)(s+3-j7)(s+3+j7)} = \frac{\mathrm{A}}{(s+5)} + \frac{\mathrm{B}}{(s+3-j7)} + \frac{B^*}{(s+3+j7)}$$

$$A = H(s)\,(s+5)\Big|_{s=-5} = \frac{(10s+156)}{\cancel{(s+5)}\left((s+3)^2+7^2\right)}\cancel{(s+5)}\Big|_{s=-5} = 2$$

$$B = H(s)(s+3-j7)\Big|_{s=-3+j7} = \frac{(10s+156)}{(s+5)\cancel{(s+3-j7)}(s+3+j7)}\cancel{(s+3-j7)}\Big|_{s=-3+j7} = -1-1\mathrm{j}$$

$$B^* = -1+1\mathrm{j}$$

$$H(s) = \frac{(10s+156)}{(s+5)\left((s+3)^2+7^2\right)} = \frac{2}{(s+5)} + \frac{-1-1\mathrm{j}}{(s+3-j7)} + \frac{-1+1\mathrm{j}}{(s+3+j7)}$$

$$a = -1; \quad b = -1; \quad \sigma = 3; \quad w = 7$$

```
>> Hs=(10*s+156)/((s+5)*(s^2+6*s+58));
>> ft = ilaplace_fga(Hs,false);
ft=@(t)(+(2*exp(-3.00000*t).*[-1.00000*cos(7.00000*t)-(-1.00000)*sin(7.00000*
t)])+(2.00000*exp(-5.00000*t)) ).*f_u(t);
```

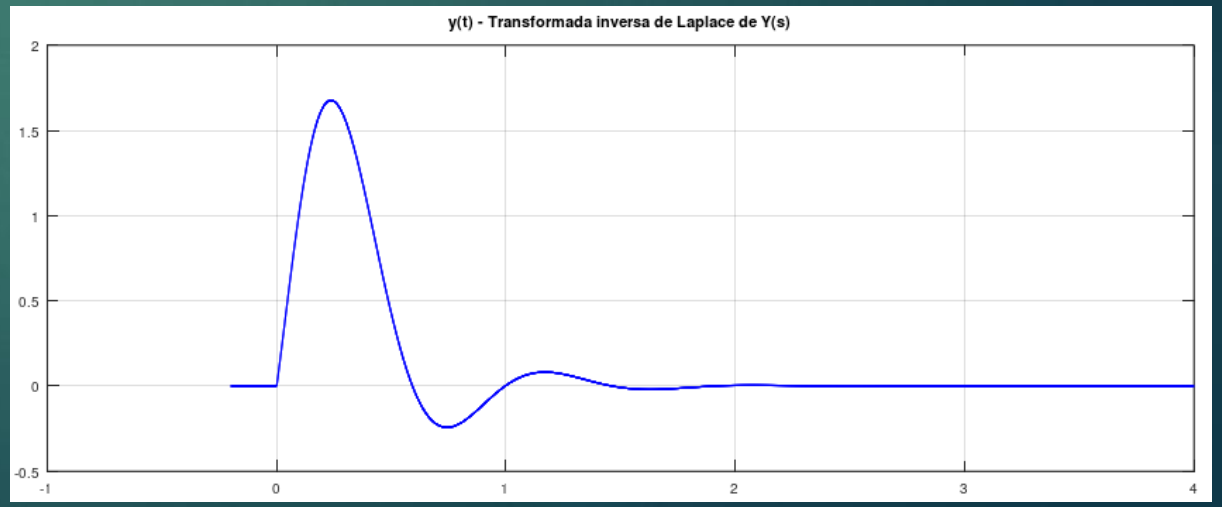
```
>> [yo,t]=Plot_Transformada_Inversa(Hs,2,'b');
ft=@(t)(+(2*(1.41421).*exp(-3.00000*t).*cos(7.00000*t+(-2.35619))
)+(2.00000*exp(-5.00000*t)) ).*f_u(t);
>>
```

Pólos complexos conjugados $-\sigma \pm jw$
com resíduos $a \pm jb$

$$\frac{a+jb}{(s+\sigma-jw)} + \frac{a-jb}{(s+\sigma+jw)} \xrightarrow{\mathcal{L}^{-1}} 2e^{-\sigma t}[\mathrm{a}\cos(wt) - b\,sen(wt)]$$

$$\frac{-1-1\mathrm{jj}}{(s+3-j7)} + \frac{-1+1\mathrm{j}}{(s+3+j7)} \xrightarrow{\mathcal{L}^{-1}} 2e^{-3t}((-1)\cos(7t) - (-1)\sin(7\mathrm{t}))$$

$$H(s) \xrightarrow{\mathcal{L}^{-1}} 2e^{-5t} + 2e^{-3t}(-\cos(7t) + \sin(7\mathrm{t}))$$



Método 3 – Combinação: Expansão em Frações Parciais + Sistema de Equações + Inspeção na Tabela

$$H(s) = \frac{(10s+156)}{(s+5)((s+3)^2+7^2)} = \frac{A}{s+5} + \frac{\mathrm{Cs+D}}{s^2+6s+58}$$

$$A = H(s)\,(s+5)\Big|_{s=-5} = \frac{(10s+156)}{(s+5)(s^2+6s+58)}(s+5)\Big|_{s=-5} = 2$$

$$\left[\frac{(10s+156)}{(s+5)(s^2+6s+58)}\right](s+5)(s^2+6s+58) = \left[\frac{2}{s+5} + \frac{\mathrm{Cs+D}}{s^2+6s+58}\right](s+5)(s^2+6s+58)$$

$$10s+156 = 2(s^2+6s+58) + (\mathrm{Cs+D})(\mathrm{s}+5)$$

$$10s+156 = (2+C)s^2 + (12+5C+D)s + (116+5D)$$

$$2+C=0 \rightarrow C=-2$$

$$116+5D=156 \rightarrow D=8$$

```
--> roots(s^2+6*s+58)
 ans  =

  -3. + 7.i
  -3. - 7.i
```

$\frac{\mathrm{Cs+D}}{(s+\sigma)^2+w^2}$	$e^{-\sigma t}\left(C\cos(wt) + \frac{D-\sigma C}{w}\,sen(wt)\right)$

$$\boxed{\begin{array}{ll} C=-2 & D=8 \\ \alpha=3 & w=7 \end{array}}$$

$$\frac{(-2)s+8}{(s+3)^2+7^2} \xRightarrow{\mathcal{L}^{-1}} e^{-3t}\left(-2\cos(7t) + \frac{8-3(-2)}{7}\,sen(7t)\right)$$

$$H(s) = \frac{2}{s+5} + \frac{-2s+8}{(s+3)^2+7^2} = 2e^{-5t} + e^{-3t}(-2\cos(7t) + 2sen(7t))$$

$$\boxed{\begin{array}{ll} C=-2 & D=8 \\ \alpha=3 & w=7 \end{array}}$$

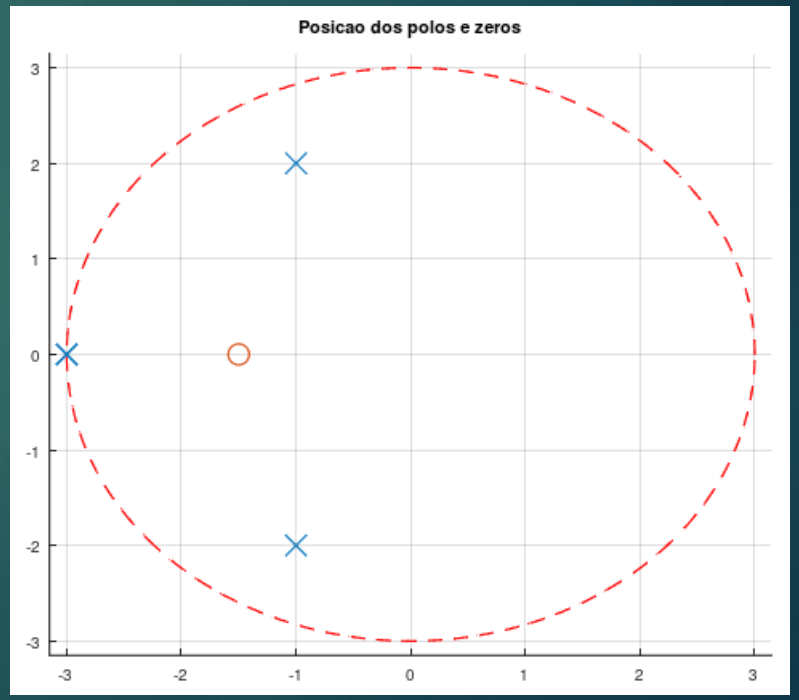
Caso 4B - Polos complexos conjugados

$$H(s) = \frac{32(6s+9)}{(s+3)^2(s^2+2s+5)} = \frac{32(6s+9)}{(s+3)^2\left((s+1)^2+2^2\right)}$$

$polos$: $-3, -3\,, -1+2j, -1-2j)$

$zero$: -1.5

```
>> Hs=32*(6*s+9)/((s+3)^2*(s^2+2*s+5));
>> plot_polos_zeros(Hs)
```



Método 1 – Expansão em Frações Parciais de H(s) e Inspeção na Tabela

$$H(s) = \frac{32(6s+9)}{(s+3)^2\left((s+1)^2+2^2\right)} = \frac{32(6s+9)}{(s+3)^2(s+1-j2)(s+1+j2)} = \frac{A}{(s+3)^2} + \frac{\mathrm{B}}{(s+3)} + \frac{\mathrm{C}}{(s+1-j2)} + \frac{C^*}{(s+1+j2)}$$

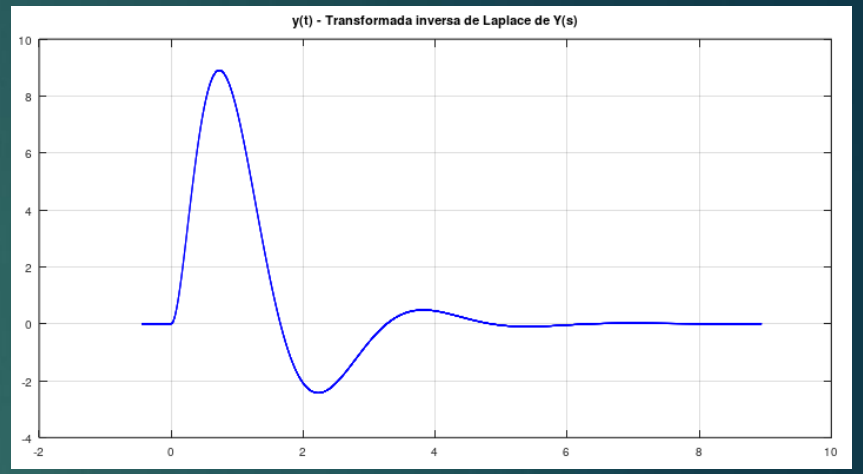
$$A = H(s)\,(s+3)^2\Big|_{s=-3} = -36$$

$$B = \frac{\partial}{\partial s}\left[H(s)\,(s+3)^2\right]\Big|_{s=-3} = 6$$

$$C = H(s)(s+1-j2)|_{s=-1+j2} = -3-12\mathrm{j} \qquad ;C^* = -3+12\mathrm{j}$$

$$H(s) = \frac{-36}{(s+3)^2} + \frac{6}{(s+3)} + \frac{-3-12\mathrm{j}}{(s+1-j2)} + \frac{-3+12\mathrm{j}}{(s+1+j2)}$$

e^{-at}	$\frac{1}{s+a}$
te^{-at}	$\frac{1}{(s+a)^2}$



Pólos complexos conjugados $-\sigma \pm jw$

com resíduos $a \pm jb$

$$\frac{a+jb}{(s+\sigma-jw)} + \frac{a-jb}{(s+\sigma+jw)} \xrightarrow{\mathcal{L}^{-1}} 2e^{-\sigma t}[\mathrm{a}\cos(wt) - b\,sen(wt)]$$

$$a = -3; \quad b = -12; \quad \sigma = 1; \quad w = 2$$

$$\frac{-3-12\mathrm{j}}{(s+1-j2)} + \frac{-3+12\mathrm{j}}{(s+1+j2)} \xrightarrow{\mathcal{L}^{-1}} = 2e^{-t}[-3\cos(2t) + 12sen(2t)]$$

$$H(s) \xrightarrow{\mathcal{L}^{-1}} h(t) = \left(e^{-3t}(-36t+6) + e^{-t}\left(-6\cos(2t) + 24sen(2t)\right)\right)\mathrm{u(t)}$$

```
>> Hs=32*(6*s+9)/((s+3)^2*(s^2+2*s+5));
>> ft = ilaplace_fga(Hs,false);
ft=@(t)(+(6.00000*exp(-3.00000*t))+(-36.00000*t.*exp(-3.00000*t)
)+(2*exp(-1.00000*t).*[-3.00000*cos(2.00000*t)-(-12.00000)*sin(2.00000*t)]) )
.*f_u(t);
```

```
>> [yo,t]=Plot_Transformada_Inversa(Hs,2,'b');
ft=@(t)(+(6.00000*exp(-3.00000*t))+(-36.00000*t.*exp(-3.00000*t)
)+(2*(12.36932).*exp(-1.00000*t).*cos(2.00000*t+(-1.81577))
) ).*f_u(t);
```

Método 3 – Combinação: Expansão em Frações Parciais + Sistema de Equações + Inspeção na Tabela

$$H(s) = \frac{32(6s+9)}{(s+3)^2\left((s+1)^2+2^2\right)} = \frac{A}{(s+3)^2} + \frac{\mathrm{B}}{(s+3)} + \frac{\mathrm{Cs+D}}{s^2+2s+5}$$

$$A = H(s)\,(s+3)^2\Big|_{s=-3} = -36$$

$$B = \frac{\partial}{\partial s}\left[H(s)\,(s+3)^2\right]\Big|_{s=-3} = 6$$

$$\left[\frac{32(6s+9)}{(s+3)^2(s^2+2s+5)}\right](s+3)^2(s^2+2s+5) = \left[\frac{-36}{(s+3)^2} + \frac{6}{(s+3)} + \frac{\mathrm{Cs+D}}{s^2+2s+5}\right](s+3)^2(s^2+2s+5)$$

$$32(6s+9) = -36(s^2+2s+5)+6(s^2+2s+5)(s+3) + (Cs+D)(s+3)^2$$
$$32(6s+9) = -36(s^2+2s+5)+6(s^2+2s+5)(s+3) + (Cs+D)(s+3)^2$$
$$192s+288 = (-36s^2-72s-180) + (6s^3+30s^2+66s+90) + (\mathrm{C}s^3 + (\mathrm{D}+6\mathrm{C})s^2 + (6\mathrm{D}+9\mathrm{C})s + 9\mathrm{D})$$
$$192s+288 = (6+\mathrm{C})s^3 + (\ldots\ldots)s^2 + (\ldots\ldots)s + (-180+90+9\mathrm{D})$$
$$(6+\mathrm{C}) = 0 \rightarrow \mathrm{C} = -6$$
$$(-180+90+9\mathrm{D}) = 288 \rightarrow \mathrm{D} = 42$$

$$\boxed{\begin{array}{c} C = -6 \\ D = 42 \end{array}}$$

$$H(s) = \frac{A}{(s+3)^2} + \frac{\mathrm{B}}{(s+3)} + \frac{\mathrm{Cs+D}}{(s+1)^2+2^2} = \frac{-36}{(s+3)^2} + \frac{6}{(s+3)} + \frac{-6s+42}{(s+1)^2+2^2}$$

$\dfrac{\mathrm{Cs+D}}{(s+\sigma)^2+w^2}$	$e^{-\sigma t}\left(C\cos(wt) + \dfrac{D-\sigma C}{w}\,sen(wt)\right)$

$$\boxed{\begin{array}{ll} C = -6 & D = 42 \\ \alpha = 1 & w = 2 \end{array}}$$

$$\frac{(-6)s+42}{(s+1)^2+2^2} \xrightarrow{\mathcal{L}^{-1}} e^{-t}\left(-6\cos(2t) + \frac{42-1(-6)}{2}\,sen(2t)\right)$$

$$H(s) \xrightarrow{\mathcal{L}^{-1}} h(t) = \left(e^{-3t}(-36t+6) + e^{-t}\left(-6\cos(2t) + 24sen(2t)\right)\right)\mathrm{u(t)}$$

```
>> Hs=32*(6*s+9)/((s+3)^2*(s^2+2*s+5));
>> disp_pfss_fga(Hs);

Transfer function 'D' from input 'u1' to output ...

         6
 y1:  -----
      s + 3

Continuous-time model.

Transfer function 'D' from input 'u1' to output ...

            -36
 y1:  -------------
      s^2 + 6 s + 9

Continuous-time model.

Transfer function 'D' from input 'u1' to output ...

         -6 s + 42
 y1:  -------------
      s^2 + 2 s + 5

Continuous-time model.
```

```
>> zero(s^2+2*s+5)
ans =

  -1 + 2i
  -1 - 2i
```